

1) Load Balancing and virtualization.

* The Technology used to distribute service request to resources is referred to as load balancing.

* Load balancing can be implemented in hardware, as is the case with F5's BIGIP servers, or in software, such as the Apache mod-proxy-balancer extension, the pound load balancer and reverse proxy software and the squid proxy and cache daemon.

* Load balancing is an optimization technique it can be used to increase utilization and throughput, lower latency, reduce response time, and avoid system overload.

* The following network resources can be load balanced.

* Network interfaces and services such as DNS, FTP and HTTP.

* connections through intelligent switches.

* processing through computer system assignment.

* Storage resources

* Without load balancing, cloud computing would be very difficult to manage. Load balancing provides the necessary redundancy to make an intrinsically unreliable system reliable through managed redirection.

* It Also provides fault tolerance when coupled with a failover mechanism. Load balancing is nearly always a feature of server farms and computer clusters and for high availability applications.

* A Load-Balancing system can use different mechanism to assigns service direction. In the simplest load-balancing mechanisms, the load balancer listens to a network port for service requests.

* When a request from a client or service requests arrives, the load balancer uses a scheduling algorithm to assign where the request is sent.

* The Algorithm can be based on a simple round robin system where the next system in a list of system gets the request. Round Robin DNS is a common application, where IP addresses are assigned out of a pool of available IP addresses.

2) Understanding Hypervisors:

* Load balancing virtualizes systems and resources by mapping a logical address to a physical address.

* If load balancing is like playing a game of hot potato, then virtual machine technologies is akin to playing slice and dice with the potato.

* Given a computer system with a certain

Set of resources, we can set aside portions of those resources to create a virtual machine.

- * From the standpoint of Applications or users, a virtual machine has all the attributes and characteristics of physical system but is strictly software that emulates a physical machine.

- * A system virtual machine has its own address space in memory, its own processor resource allocation and its own device I/O using its own virtual device drivers. Some virtual machines are designed to run only a single application or process and are referred to as process virtual machines.

- * A virtual machine is a computer that is walled off from the physical computer than the virtual machine is running on.

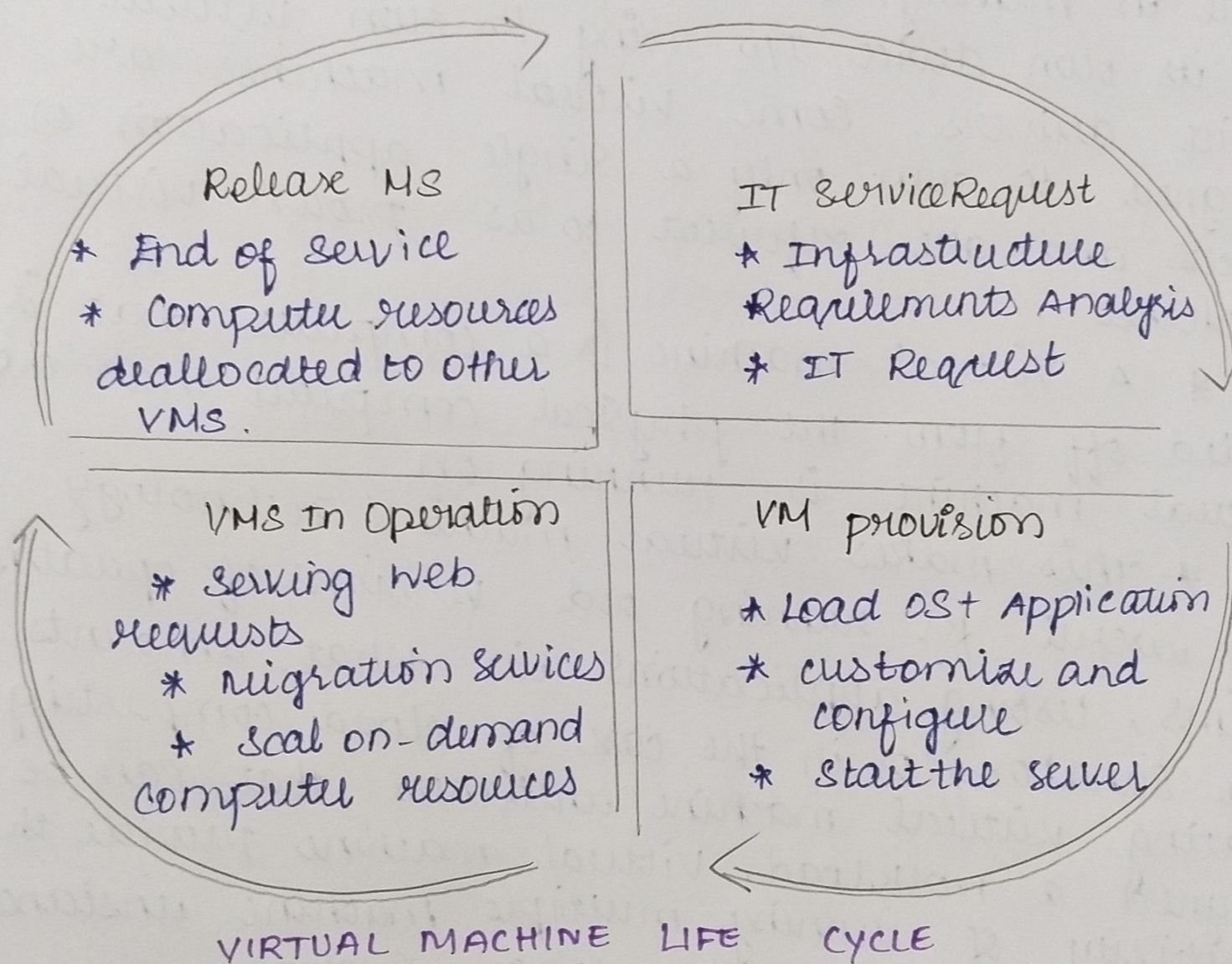
- * This makes virtual machine technology very useful for running old versions of operating systems, testing applications in what amounts to a sandbox, or in the case of cloud computing, creating virtual machine instances that can be assigned a workload. Virtual machines provide the capability of running multiple machine instances, each with their own operating system.

- * The downside of virtual machine technologies is that having resources indirectly addressed means there is some level of overhead.

- * From the standpoint of cloud computing, these features enable VMs to manage application provisioning, provide for machine instance cloning.

3) virtual Machines provisioning and Manageability

* The Typical Life cycle of VM and its Major possible states of operation, which make the management and automation of VMs in virtual and cloud environments easier than in traditional computing environments.



* Steps to provision VM

* You need to select a server from a pool of available servers along with the appropriate OS template you need to provision the virtual machine.

* Next, you need to load the appropriate software in the previous step, device drivers, middleware and the needed applications for the service required.

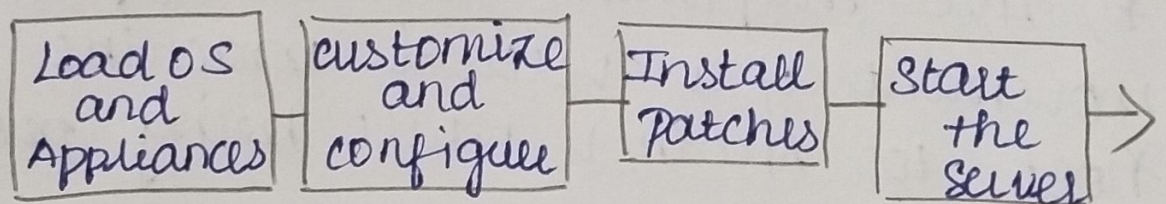
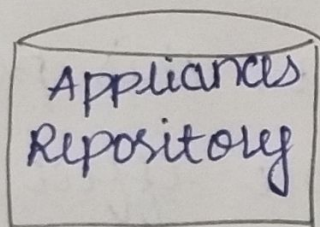
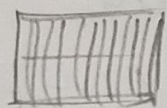
* Thirdly, you need to customize and configure the machine to configure an associated network and storage resources.

* Finally, the virtual server is ready to start with its newly loaded software.

* Typically, these are the tasks required or being performed by an IT or a data center's specialist to provision a particular virtual machine.

* Normally, virtual machine can be provisioned by manually installing an operating system, by using a preconfigured VM template, by cloning an existing VM, or by importing a physical server or a virtual server from another hosting platform.

servers pool



virtual machine
provision process.

* This enables the Administrator to quickly provision a correctly configured virtual server on demand.

Virtual Machine Migration Services:

* Migration service, in the context of virtual machines, is the process of moving a virtual machine from one host server or storage location to another.

* There are different techniques of VM migration, hot/live migration, cold/regular migration, and live storage migration of a virtual machine.

* In this process, all key machines' components, such as CPU, storage disks, network, and memory, are completely virtualized, thereby facilitating the entire state of a virtual machine to be captured by a set of easily moved data files.

migrations Techniques:

* Live Migration

* High Availability.

* Live migration can be defined as the movement of a virtual machine from one physical host to another while being powered on.

* When it is properly carried out, this process takes place without any noticeable effect from the end user's point of view.

* Live migration can also be used for load balancing in which work is shared among computers in order to optimize the utilization of available CPU service resources.